

# Econometrics Assignment 2

## Panel Data Econometrics

### Assignment Overview

This assignment introduces panel data econometrics using real-world country-year datasets.

Panel data combines:

- cross-sectional variation across entities
- time-series variation within each entity

In this assignment, entities are countries and time periods are years.

The core goal is to understand how panel methods differ from ordinary cross-sectional regression.

# Topics to Learn Before Attempting This Assignment

## A. Panel Data Structure

- entity-year datasets
- balanced vs unbalanced panels
- within-entity variation
- between-entity variation
- panel indexing

## B. Pooled OLS

- pooled regression intuition
- assumptions behind pooled OLS
- why pooled OLS can be misleading
- omitted entity-level heterogeneity

## C. Fixed Effects

- entity fixed effects
- time fixed effects
- two-way fixed effects
- within transformation intuition
- time-invariant omitted variables

## D. Random Effects

- random effects intuition
- FE vs RE distinction
- correlation between entity effects and regressors
- when RE is plausible
- why RE assumptions are often strong

## **E. Panel Inference**

- clustered standard errors
- serial correlation within entities
- heteroskedasticity in panel data
- country-level clustering
- two-way clustering intuition

## **F. Dynamic Panels**

- lagged dependent variables
- persistence
- dynamic panel bias
- Nickell bias intuition
- Arellano-Bond motivation

## **G. Panel Difference-in-Differences**

- treatment and control groups
- pre-period and post-period
- treatment timing
- parallel trends
- panel DiD intuition

# Required Submission

Submit:

1. One Jupyter notebook
2. One short report

Your report must include:

- research question
- dataset description
- panel construction
- pooled OLS results
- fixed effects results
- random effects results
- clustered standard errors
- dynamic panel discussion
- final economic interpretation
- clear distinction between association and causation

Use:

- statsmodels
- linearmodels
- pandas
- numpy
- matplotlib or seaborn

# Problem 1 — Country Growth and Inflation Panel

## Dataset

Use the **World Bank World Development Indicators**.

Use annual country-level data.

Suggested sample:

- Countries: India, China, Bangladesh, Vietnam, Indonesia, Brazil, Mexico, South Africa, Turkey, Philippines
- Years: 2000–2023

You may expand the country list if desired.

## Research Question

**Does inflation have a systematic relationship with economic growth across countries over time?**

This is not automatically causal. You are estimating conditional associations.

## Variables

| Concept                | Suggested WDI Code | Variable Name |
|------------------------|--------------------|---------------|
| GDP growth             | NY.GDP.MKTP.KD.ZG  | gdp_growth    |
| Inflation              | FP.CPI.TOTL.ZG     | inflation     |
| Trade openness         | NE.TRD.GNFS.ZS     | trade_open    |
| Investment share       | NE.GDI.TOTL.ZS     | investment    |
| Population growth      | SP.POP.GROW        | pop_growth    |
| Government expenditure | NE.CON.GOVT.ZS     | gov_exp       |

## Dependent Variable

- $gdp\_growth_{it}$

## Main Independent Variable

- $inflation_{it}$

## Controls

- $trade\_open_{it}$
- $investment_{it}$
- $pop\_growth_{it}$
- $gov\_exp_{it}$

## Part A — Construct the Panel

Create a country-year dataset.

Set the panel index as:

```
df = df.set_index(["country", "year"])
```

### Tasks

1. Download or import the WDI data.
2. Keep the required countries and years.
3. Rename variables clearly.
4. Set the country-year panel index.
5. Check missing values.

### Questions

1. How many countries are included?
2. How many years are included?
3. Is the panel balanced?
4. Which variables have missing values?
5. Which countries have the most missing data?

## Part B — Pooled OLS

Estimate:

$$gdp\_growth_{it} = \beta_0 + \beta_1 inflation_{it} + \beta_2 investment_{it} + \beta_3 trade\_open_{it} + \beta_4 pop\_growth_{it} + u_{it}$$

Python skeleton:

```
pooled = smf.ols(
    "gdp_growth ~ inflation + investment + trade_open + pop_growth",
    data=df_reset
).fit()

print(pooled.summary())
```

### Questions

1. Interpret  $\beta_1$ .
2. Is inflation statistically significant?
3. What does the pooled model assume about countries?
4. Why might this assumption be unrealistic?

**Expected issue:** pooled OLS ignores that countries may have different long-run growth levels due to institutions, geography, demographics, industrial structure, and policy regimes.



## Part C — Entity Fixed Effects

Estimate:

$$gdp\_growth_{it} = \beta_1 inflation_{it} + \beta_2 investment_{it} + \beta_3 trade\_open_{it} + \beta_4 pop\_growth_{it} + \alpha_i + u_{it}$$

where  $\alpha_i$  is the country fixed effect.

Python skeleton:

```
from linearmodels.panel import PanelOLS

fe_entity = PanelOLS.from_formula(
    "gdp_growth ~ inflation + investment + trade_open
    + pop_growth + EntityEffects",
    data=df
).fit()

print(fe_entity.summary)
```

### Questions

1. What changes when country fixed effects are added?
2. Does the inflation coefficient become larger, smaller, or less significant?
3. What country-level omitted variables are absorbed by entity fixed effects?
4. Why can you not estimate coefficients on time-invariant country traits in an entity FE model?

Examples of absorbed traits:

- geography
- colonial history
- deep institutional quality
- long-run culture
- natural resource endowment
- baseline industrial structure

## Part D — Entity Fixed Effects and Time Fixed Effects

Estimate:

$$gdp\_growth_{it} = \beta_1 inflation_{it} + \beta_2 investment_{it} + \beta_3 trade\_open_{it} + \beta_4 pop\_growth_{it} + \alpha_i + \lambda_t + u_{it}$$

where:

- $\alpha_i$  is the country fixed effect
- $\lambda_t$  is the year fixed effect

Python skeleton:

```
fe_tw = PanelOLS.from_formula(
    "gdp_growth ~ inflation + investment + trade_open
    + pop_growth + EntityEffects + TimeEffects",
    data=df
).fit()

print(fe_tw.summary)
```

### Questions

1. What global shocks are captured by time fixed effects?
2. Does the inflation coefficient survive after adding time fixed effects?
3. Compare  $R^2$  across pooled OLS, entity FE, and two-way FE.
4. Which model is most credible? Why?

Examples of time shocks:

- 2008 global financial crisis
- COVID-19 shock
- commodity price cycles
- global interest-rate conditions
- global trade slowdown

## Part E — Clustered Standard Errors

Re-estimate the two-way FE model with standard errors clustered by country:

```
fe_tw_clustered = PanelOLS.from_formula(
    "gdp_growth ~ inflation + investment + trade_open
    + pop_growth + EntityEffects + TimeEffects",
    data=df
).fit(
    cov_type="clustered",
    cluster_entity=True
)

print(fe_tw_clustered.summary)
```

### Questions

1. How do standard errors change after clustering?
2. Why might errors be correlated within countries over time?
3. Does your conclusion about inflation change?
4. Why are clustered standard errors usually preferred in panel data?

**Key concept:** observations from the same country are usually not independent. India in 2001 and India in 2002 are more related than India in 2001 and Brazil in 2001.

## Part F — Random Effects

Estimate:

$$gdp\_growth_{it} = \beta_0 + \beta_1 inflation_{it} + \beta_2 investment_{it} + \beta_3 trade\_open_{it} + \beta_4 pop\_growth_{it} + c_i + u_{it}$$

Python skeleton:

```
from linearmodels.panel import RandomEffects

re = RandomEffects.from_formula(
    "gdp_growth ~ inflation + investment + trade_open + pop_growth",
    data=df
).fit()

print(re.summary)
```

### Questions

1. What does random effects assume about country-specific effects?
2. Why is this assumption strong?
3. Compare the inflation coefficient under FE and RE.
4. Which model would you trust more? Why?

**Core distinction:** fixed effects allows country effects to be correlated with regressors. Random effects assumes country effects are uncorrelated with regressors.

## Part G — Dynamic Panel Intuition

Create lagged GDP growth:

```
df["gdp_growth_lag1"] = df.groupby(level=0)["gdp_growth"].shift(1)
```

Estimate:

$$gdp\_growth_{it} = \rho gdp\_growth_{i,t-1} + \beta_1 inflation_{it} + \beta_2 investment_{it} + \beta_3 trade\_open_{it} + \alpha_i + \lambda_t + u_{it}$$

Python skeleton:

```
dyn_fe = PanelOLS.from_formula(
    "gdp_growth ~ gdp_growth_lag1 + inflation + investment
    + trade_open + pop_growth + EntityEffects + TimeEffects",
    data=df.dropna()
).fit(
    cov_type="clustered",
    cluster_entity=True
)

print(dyn_fe.summary)
```

### Questions

1. Is lagged growth statistically significant?
2. Does growth show persistence?
3. What happens to the inflation coefficient after adding lagged growth?
4. Why can lagged dependent variables create bias in fixed-effects panels?
5. Why does the Arellano-Bond estimator exist?

**Important:** including lagged dependent variables in a fixed-effects model can create dynamic panel bias, especially when  $T$  is small.

## Final Write-Up for Problem 1

Write a 700–900 word report answering:

Is inflation negatively associated with growth once country effects, year shocks, and persistence are considered?

Your conclusion must distinguish:

- pooled association
- within-country association
- time-adjusted association
- causal claim

## Problem 2 — CO<sub>2</sub> Emissions, Energy Use, and Economic Development

### Dataset

Use the **Our World in Data CO<sub>2</sub> and Greenhouse Gas Emissions Dataset**.

The dataset follows a one-row-per-location-per-year structure, making it suitable for panel analysis.

Suggested sample:

- Countries: India, China, United States, Germany, Japan, Brazil, Indonesia, Vietnam, South Africa, Mexico
- Years: 1990–2023

### Research Question

**As countries become richer, do their CO<sub>2</sub> emissions per person rise or fall?**

This problem is useful for panel data because countries differ strongly in:

- geography
- industrial structure
- energy systems
- development stage
- environmental policy

### Variables

| Concept                    | Suggested Variable |
|----------------------------|--------------------|
| CO <sub>2</sub> per capita | co2_per_capita     |
| GDP per capita             | gdp / population   |
| Population                 | population         |
| Energy per capita          | energy_per_capita  |
| Country                    | country            |
| Year                       | year               |

Create:

```
df["log_co2_pc"] = np.log(df["co2_per_capita"])  
df["log_gdp_pc"] = np.log(df["gdp"] / df["population"])  
df["log_pop"] = np.log(df["population"])
```

Drop rows with missing or non-positive values before taking logs.



## Part A — Build the Panel

### Tasks

1. Import the OWID CO<sub>2</sub> dataset.
2. Keep the selected countries and years.
3. Create log variables.
4. Set the country-year panel index.
5. Plot CO<sub>2</sub> per capita over time for four countries.

Suggested countries for plots:

- India
- China
- United States
- Germany

### Questions

1. How many countries and years are in your sample?
2. Is the panel balanced?
3. Which countries have the highest average CO<sub>2</sub> per capita?
4. Which countries have the lowest average CO<sub>2</sub> per capita?
5. What broad time trends do you observe?

## Part B — Pooled OLS

Estimate:

$$\log\_co2\_pc_{it} = \beta_0 + \beta_1 \log\_gdp\_pc_{it} + \beta_2 \log\_pop_{it} + u_{it}$$

Python skeleton:

```
pooled = smf.ols(  
    "log_co2_pc ~ log_gdp_pc + log_pop",  
    data=df_reset  
)  
.fit()  
  
print(pooled.summary())
```

### Questions

1. Interpret  $\beta_1$  as an elasticity.
2. Does richer country-year data correlate with higher emissions per person?
3. Why might pooled OLS be misleading?
4. What unobserved country traits may bias the estimate?

Possible omitted variables:

- energy mix
- climate
- industrial structure
- urbanization
- policy regime
- public transport systems
- natural resource endowment

## Part C — Entity Fixed Effects

Estimate:

$$\log\_co2\_pc_{it} = \beta_1 \log\_gdp\_pc_{it} + \beta_2 \log\_pop_{it} + \alpha_i + u_{it}$$

Python skeleton:

```
fe_entity = PanelOLS.from_formula(
    "log_co2_pc ~ log_gdp_pc + log_pop + EntityEffects",
    data=df
).fit()

print(fe_entity.summary)
```

### Questions

1. How does  $\beta_1$  change compared with pooled OLS?
2. What does  $\beta_1$  mean now?
3. Why is this a within-country estimate?
4. What country-level factors are removed?

**Interpretation:** with entity fixed effects,  $\beta_1$  compares a country to itself over time.

## Part D — Time Fixed Effects

Estimate:

$$\log\_co2\_pc_{it} = \beta_1 \log\_gdp\_pc_{it} + \beta_2 \log\_pop_{it} + \alpha_i + \lambda_t + u_{it}$$

Python skeleton:

```
fe_tw = PanelOLS.from_formula(  
    "log_co2_pc ~ log_gdp_pc + log_pop  
    + EntityEffects + TimeEffects",  
    data=df  
)
```

```
print(fe_tw.summary)
```

### Questions

1. What does time fixed effects control for?
2. Does the GDP-emissions relationship remain after controlling for global year shocks?
3. How do time fixed effects affect the interpretation?
4. Which model gives the most defensible estimate?

## Part E — Clustered Standard Errors

Estimate:

```
fe_tw_clustered = PanelOLS.from_formula(
    "log_co2_pc ~ log_gdp_pc + log_pop
    + EntityEffects + TimeEffects",
    data=df
).fit(
    cov_type="clustered",
    cluster_entity=True
)

print(fe_tw_clustered.summary)
```

### Questions

1. Compare normal standard errors and clustered standard errors.
2. Does statistical significance change?
3. Why might CO<sub>2</sub> errors be serially correlated within countries?
4. Should you cluster by country, year, or both? Explain.

**Good answer:** cluster by country when the concern is serial correlation within countries. Two-way clustering may be useful if shocks are also correlated across countries in the same year.

## Part F — Random Effects

Estimate:

$$\log\_co2\_pc_{it} = \beta_0 + \beta_1 \log\_gdp\_pc_{it} + \beta_2 \log\_pop_{it} + c_i + u_{it}$$

Python skeleton:

```
re = RandomEffects.from_formula(  
    "log_co2_pc ~ log_gdp_pc + log_pop",  
    data=df  
)  
.fit()  
  
print(re.summary)
```

### Questions

1. What assumption makes random effects more efficient than fixed effects?
2. Why might that assumption fail here?
3. Compare RE with FE.
4. Would you trust RE for this research question?

**Expected issue:** country-specific traits such as industrial structure, fossil fuel reserves, climate, and institutions are likely correlated with GDP per capita.

## Part G — Dynamic Panel

Create lagged emissions:

```
df["log_co2_pc_lag1"] = df.groupby(level=0)["log_co2_pc"].shift(1)
```

Estimate:

$$\log\_co2\_pc_{it} = \rho \log\_co2\_pc_{i,t-1} + \beta_1 \log\_gdp\_pc_{it} + \beta_2 \log\_pop_{it} + \alpha_i + \lambda_t + u_{it}$$

Python skeleton:

```
dyn = PanelOLS.from_formula(
    "log_co2_pc ~ log_co2_pc_lag1 + log_gdp_pc
    + log_pop + EntityEffects + TimeEffects",
    data=df.dropna()
).fit(
    cov_type="clustered",
    cluster_entity=True
)

print(dyn.summary)
```

### Questions

1. Is CO<sub>2</sub> per capita persistent?
2. Does adding lagged emissions reduce the GDP coefficient?
3. Why are emissions likely dynamic?
4. Why should you be cautious about interpreting this model?

**Expected intuition:** emissions are persistent because energy systems, factories, transport networks, and household behavior change slowly.

## Optional Extension — Environmental Kuznets Curve

Estimate:

$$\log_{co2\_pc_{it}} = \beta_1 \log_{gdp\_pc_{it}} + \beta_2 (\log_{gdp\_pc_{it}})^2 + \alpha_i + \lambda_t + u_{it}$$

Create:

```
df["log_gdp_pc_sq"] = df["log_gdp_pc"] ** 2
```

### Questions

1. Is the linear term positive and the squared term negative?
2. Does the data suggest emissions rise and then fall with income?
3. Is this result robust to entity FE and time FE?
4. Why should you be cautious about claiming an Environmental Kuznets Curve?



## Marking Scheme

| Component                            | Marks |
|--------------------------------------|-------|
| Panel construction and data cleaning | 15    |
| Pooled OLS models                    | 10    |
| Entity fixed effects                 | 15    |
| Time fixed effects                   | 15    |
| Clustered standard errors            | 10    |
| Random effects comparison            | 10    |
| Dynamic panel discussion             | 10    |
| Interpretation and written report    | 15    |
| Total                                | 100   |

# **What You Should Learn**

By the end of this assignment, you should understand the following distinctions:

## **Pooled OLS**

Pooled OLS compares countries as if they are ordinary independent observations.

## **Entity Fixed Effects**

Entity FE compares each country against itself over time.

## **Time Fixed Effects**

Time FE controls for global shocks common to all countries in a given year.

## **Random Effects**

Random effects uses both within-country and between-country variation, but assumes country effects are uncorrelated with regressors.

## **Clustered Standard Errors**

Clustered standard errors change inference, not coefficient estimates.

## **Dynamic Panel Models**

Dynamic panel models add persistence, but create bias concerns when lagged dependent variables are combined with fixed effects.

## **Panel Difference-in-Differences**

Panel DiD can be used when treatment timing differs across entities and the goal is to compare treated and untreated units before and after a policy change.

## Final Reminder

Your conclusions must be cautious.

Most models in this assignment estimate associations, not causal effects.

Do not write that inflation causes growth changes or GDP causes emissions changes unless you have a valid identification strategy.